

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460887.956 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.05
Transit depth (Rp/Rs)^2	2.89 +/- 1.71 [%]
Orbital Inclination (inc)	84.9 +/- 2.89
Airmass coefficient 1 (a1)	0.031 +/- 0.013
Airmass coefficient 2 (a2)	2.24 +/- 0.26
Scatter in the residuals of the lightcurve fit is	40.03 %
Best Comparison Star	#2 - [392, 242]
Optimal Aperture	2.29
Optimal Annulus	6.5
Transit Duration (day)	0.061 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.05 vs 0.1618 (deviation 0.06σ)
Duration fit vs published	0.061 d vs 0.0721 d (deviation 0.11σ)
Mid-time O–C	34.1 min
Depth SNR	1.2
Residual scatter	40.03 %

Light curve

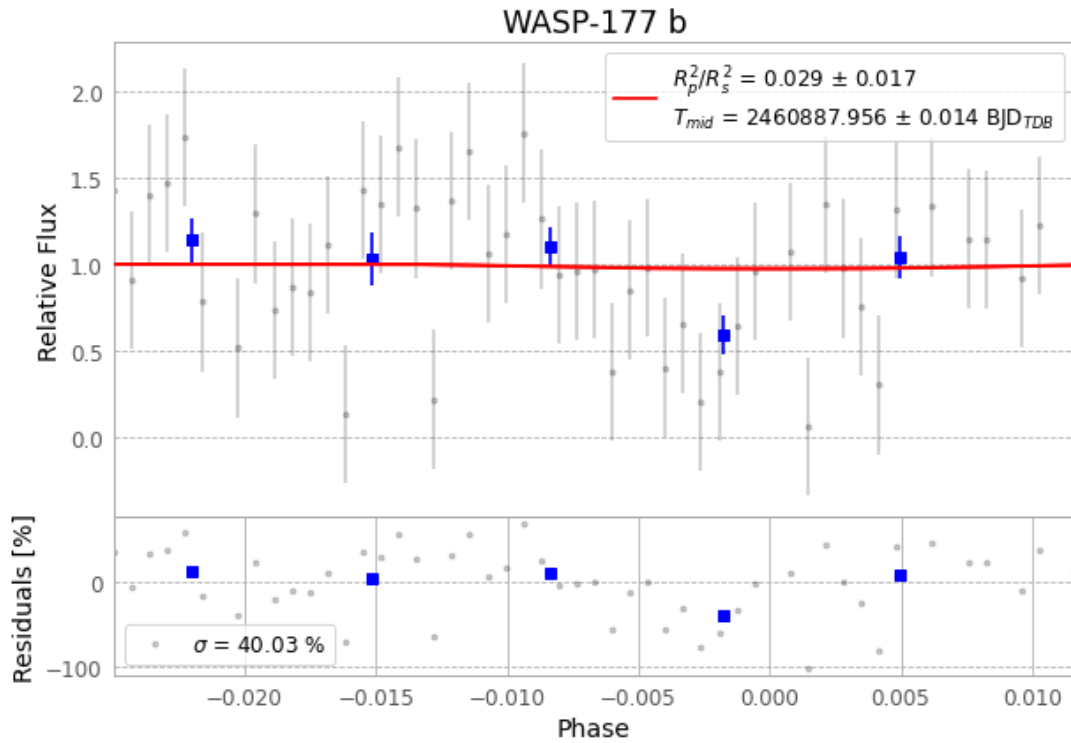


Figure 1 — FinalLightCurve_WASP-177 b_2025-07-31.png (SHA-256 dab00771127af38...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 27], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1fc147d73f74fad0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), WASP-177250731085715.FITS → WASP-177250731113916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-250731.log (0ecd6f8b9de7...), FinalLightCurve_WASP-177 b_2025-07-31.png (dab00771127...), FinalLightCurve_WASP-177 b_2025-07-31.csv (4b9ac66cfcd7...)

Compute resources

Wall time 148.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:12Z.